SEARCH · RELEVANCE · 2026-06-05 · X² P = 0.3066

Sample ratio passes.

The observed split is close enough to the expected split that the rest of the analysis can proceed. This gate fires before any metric value is read.

WHAT THIS GATE DOES

Proceed to metrics. No assignment investigation required.

A sample-ratio mismatch invalidates everything downstream — observed lift could just be a different population in each arm. The gate fires before any metric is read (R2).

OBSERVED SPLIT

5944 / 6056

expected 6000 / 6000

X² STATISTIC	P-VALUE
1.05	0.3066
THRESHOLD	**VERDICT**
0.0005	PASS

READING THIS CHECK

A small p-value here invalidates the rest of the analysis.

χ^2 compares the observed split against the expected split. A p-value above the threshold means the split is consistent with random allocation. The analyzer is unblocked and reads the primary metric next.

COMPUTATION TRACE

srm_check is the only path to this number.

All SRM evaluations run through `agentxp.stats.srm.srm_check`; the function returns the trace below verbatim. Any change to the threshold or formula must come from a code change, not an ad-hoc call site (R4).

TEST	srm_chi_squared
OBSERVED	[5944, 6056]
EXPECTED	[6000, 6000]
CHI2_STAT	1.0453
P_VALUE	0.306584
THRESHOLD	0.0005

DF	1
VERDICT	PASS
FORMULA	chi^2 = sum((obs − exp)^2 / exp) · p from chi2(df=k−1)
